

CS305 - Quiz - 2026

Max Marks: 20 Marks

Max Time: 40 mins

Set A

Date: 30 August 2026

Name: _____

Roll Number: _____

Instructions:

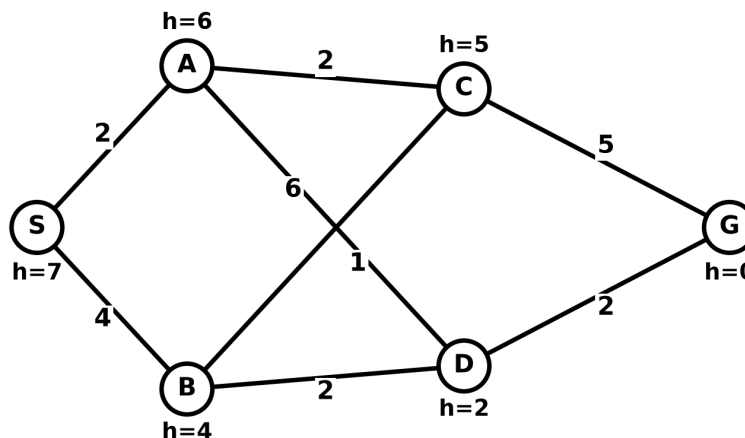
- This is a closed-notes and closed-book quiz.
- Section A has 15 multiple-choice questions and Section B has 5 fill-in-the-blank questions. Each question carries 1 mark.
- Only one option is correct in each Section A question.
- Each Section A question includes a “None of these” choice; its letter may vary across randomized sets.
- Question order is randomized within each section, and the answer-choice order is randomized in Section A.
- For graph, tree, and game-tree questions, use the traversal and tie-breaking order stated in that question.
- Write each Section B answer in the blank provided.
- Please do not use smart, connected, or AI-enabled devices.
- Copying or cheating in any form is strictly prohibited.

Best of luck!

Section A - Multiple-Choice Questions

Q1. Run A* graph search from S to G using $f(n) = g(n) + h(n)$. If a cheaper path to a frontier node is found, update its g-value. Stop when G is removed from the frontier; on equal f-values, select the alphabetically earlier node. What path cost is returned?

Edge costs and heuristic values are given in the figure and in this equivalent text specification: edges S-A:2, S-B:4, A-C:2, A-D:6, B-C:1, B-D:2, C-G:5, D-G:2; heuristics $h(S)=7$, $h(A)=6$, $h(B)=4$, $h(C)=5$, $h(D)=2$, $h(G)=0$.



- a. None of these
- b. 15
- c. 10
- d. 9
- e. 8

Q2. A proof procedure never derives a sentence unless it is entailed by the knowledge base. However, for some knowledge base KB and sentence α , $KB \models \alpha$ but the procedure cannot derive α . Which description is correct?

- a. Complete but not sound
- b. Neither sound nor complete
- c. Both sound and complete
- d. None of these
- e. Sound but not complete

Q3. A tree has the following left-to-right child lists: S:(A,B); A:(C,D); C:(E,F); E:(G). Nodes B, D, and F are non-goal leaves; G is the only goal. Depth-first search stops at the goal. What is the visit sequence up to and

including G?

Explore children left to right. Test each node when it is first visited.

- None of these
- S, A, D, C, G
- S, A, C, E, G
- S, A, C, F, G
- S, B, A, C, G

Q4. Resolve the clauses $(P \vee Q \vee \neg R)$ and $(\neg P \vee S)$ on P. Which resolvent is obtained?

- a. $Q \wedge \neg R \wedge S$
- b. $Q \vee R \vee S$
- c. None of these
- d. $Q \vee \neg R \vee S$
- e. $P \vee Q \vee S$

Q5. An office-cleaning robot receives only the doorway sensor reading available at the current instant. It stores which rooms it has already visited and applies condition-action rules to that stored state; it does not plan ahead or compare future outcomes. Which agent type is described?

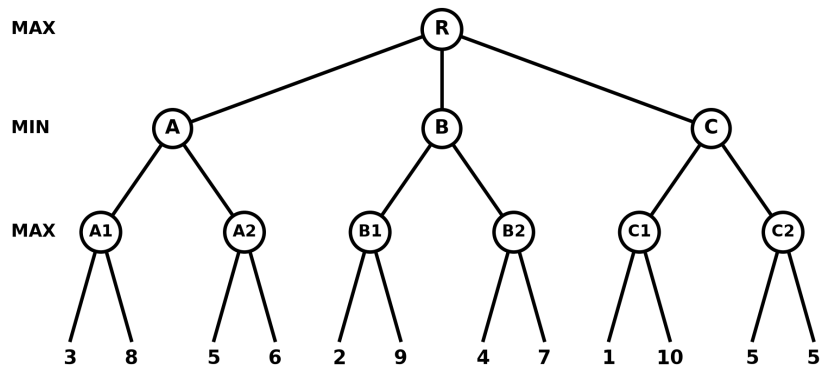
- a. Goal-based agent
- b. None of these
- c. Utility-based agent
- d. Model-based reflex agent
- e. Simple reflex agent

Q6. For assignments $(P,Q) = (T,T), (T,F), (F,T), (F,F)$ in that order, what is the truth-value sequence of $((P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg P))$?

- a. T, T, F, T
- b. T, T, T, T
- c. T, F, T, T
- d. F, F, F, F
- e. None of these

Q7. In the game tree, levels alternate as MAX (R) $\rightarrow$ MIN (A,B,C) $\rightarrow$ MAX (A1,A2,B1,B2,C1,C2) $\rightarrow$ terminal utilities. Children are evaluated left to right. What is the minimax value at R?

The equivalent terminal specification is A1:(3,8), A2:(5,6), B1:(2,9), B2:(4,7), C1:(1,10), C2:(5,5).



- 7
- 5
- None of these
- 6
- 10

Q8. A robot moves on a grid from a specified start cell to a specified destination. It may move one cell north, south, east, or west when the adjacent cell is not a wall. Which pair correctly identifies an action and the transition model?

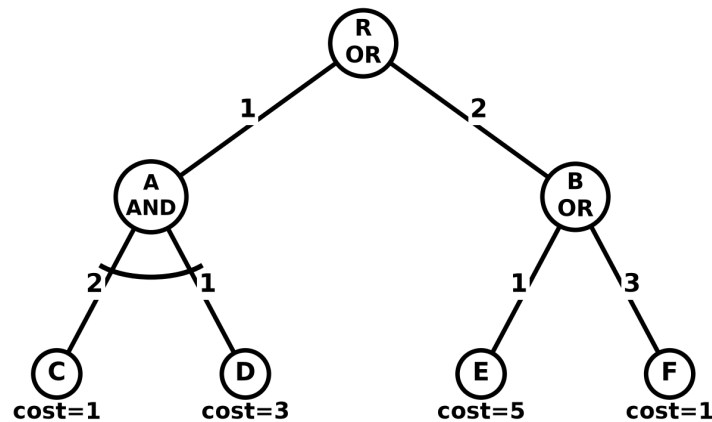
- a. Action: estimate distance to the destination; transition model: return the heuristic at the current cell
- b. Action: choose a legal direction; transition model: return the adjacent free cell reached
- c. Action: name the destination cell; transition model: test whether the destination was reached
- d. Action: occupy the current cell; transition model: return the set of legal directions
- e. None of these

Q9. In the standard Turing Test (not the Total Turing Test), an evaluator exchanges typed messages with one human and one machine without seeing either. Under the criterion taught in the course, when does the machine pass?

- a. When the machine gives factually correct answers in every exchange
- b. When the machine's internal reasoning exactly duplicates human reasoning
- c. None of these
- d. When the evaluator cannot reliably distinguish the machine from the human from their conversation
- e. When the machine recognizes physical objects and manipulates them correctly

Q10. For the AND-OR tree below, a leaf's number is its terminal cost. The cost of using an edge is added to the child cost. At an AND node, add the costs of all required children; at an OR node, take the minimum alternative. What is the backed-up cost at R?

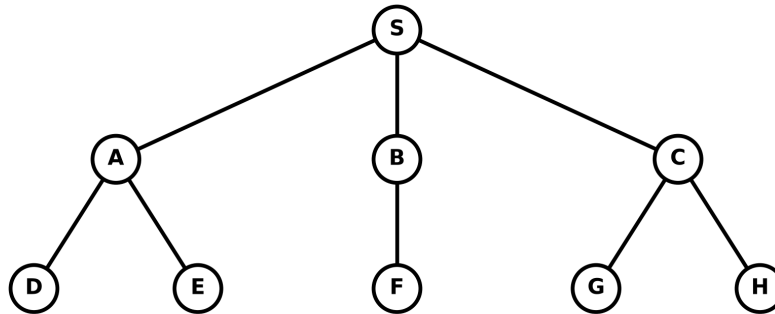
The equivalent text specification is: R is OR with R-A:1 and R-B:2; A is AND with A-C:2 and A-D:1; B is OR with B-E:1 and B-F:3; terminal costs are C=1, D=3, E=5, F=1.



- a. 6
- b. None of these
- c. 4
- d. 8
- e. 14

Q11. Using breadth-first tree search, what is the traversal from S?
Enqueue children left to right.

The figure is equivalently specified by the following left-to-right child lists: S:(A,B,C); A:(D,E); B:(F); C:(G,H).



- a. None of these
- b. S, A, B, C, E, D, F, H, G
- c. S, C, B, A, H, G, F, E, D
- d. S, A, D, E, B, F, C, G, H
- e. S, A, B, C, D, E, F, G, H

Q12. A scheduling assistant first filters to meeting times at which every required participant can attend. Among those feasible times, it computes a numerical utility score from delay, participant preferences, and room cost, then chooses the highest-utility time. Which agent type best describes this final decision rule?

- a. Utility-based agent
- b. Goal-based agent
- c. Simple reflex agent
- d. Model-based reflex agent
- e. None of these

Q13. The domain contains exactly two students, Anu and Bala, and exactly two courses, AI and Logic. Anu takes AI but not Logic; Bala takes both AI and Logic. Let Student(x), Course(y), and Takes(x,y) have their usual meanings. Which formula is true in this model?

- a. None of these
- b. $\forall y (Course(y) \rightarrow \forall x (Student(x) \rightarrow Takes(x,y)))$
- c. $\exists x (Student(x) \wedge \forall y (Course(y) \rightarrow Takes(x,y)))$
- d. $\forall x (Student(x) \rightarrow \forall y (Course(y) \rightarrow Takes(x,y)))$
- e. $\exists y (Course(y) \wedge \forall x (Student(x) \rightarrow \neg Takes(x,y)))$

Q14. Let Student(x) mean “x is a student” and Prepared(x) mean “x is prepared.” Which formula represents “Every student is prepared”?

- a. $\exists x (Student(x) \rightarrow Prepared(x))$
- b. $\forall x (Student(x) \rightarrow Prepared(x))$
- c. $\forall x (Student(x) \wedge Prepared(x))$
- d. None of these
- e. $\exists x (Student(x) \wedge Prepared(x))$

Q15. After expanding node A, an A* search has the following frontier. It uses $f(n) = g(n) + h(n)$ and, on equal f-values, selects the alphabetically earlier node.

B: g=4, h=3; C: g=2, h=6; D: g=5, h=3.

Which node is selected next?

- a. None of these
- b. D
- c. B
- d. A
- e. C

Section B - Fill in the Blanks

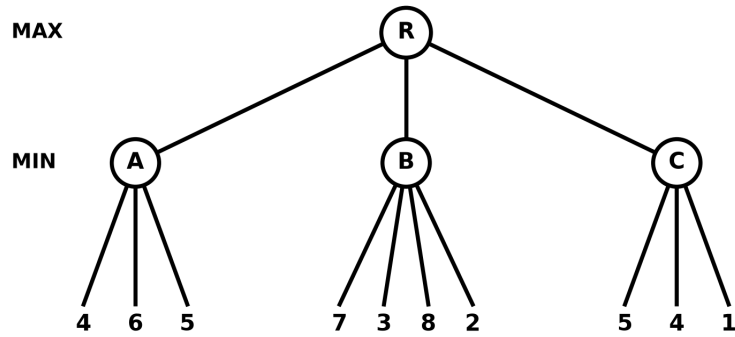
Write one number or a term of at most three words in each blank.

Q16. From $(P \rightarrow Q)$ and $\neg Q$, the inference rule that derives $\neg P$ is called _____.

Q17. Given the clauses $(P \vee Q)$, $\neg P$, and $\neg Q$, a resolution refutation terminates by deriving the _____.

Q18. Apply standard alpha-beta pruning left to right to a MAX root with three MIN children A, B, and C. A cutoff occurs when $\alpha \geq \beta$. The number of terminal leaves that are pruned is _____.

The equivalent terminal specification is A:(4,6,5); B:(7,3,8,2); C:(5,4,1).



Q19. An A* graph search has recorded $g(X)=11$. It discovers a new path to X through Y with $g(Y)=4$ and edge cost $c(Y,X)=2$. If $h(X)=3$, the updated value $f(X)$ is _____.

Q20. A formal system that derives both R and $\neg R$ lacks the property of _____.